Numerical Analysis I

Quiz 8B Name:_____

Question 1.

Consider the function

$$f(x) = \sqrt{x}, \qquad x > 0.$$

We want to approximate $f(x)$ using polynomial interpolation at the four nodes

$$x_0 = 1, \qquad x_1 = 4, \qquad x_2 = 9, \qquad x_3 = 16.$$

1. Here is the Newton divided difference table.

x_i	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, x_{i+1}, x_{i+2}, x_{i+3}]$
1	1	$\frac{1}{3}$	$-\frac{1}{60}$	$\frac{1}{1260}$
4	2	$\frac{1}{5}$	$-\frac{1}{210}$	
9	3	$\frac{1}{7}$		
16	4			

   Explain how the values $\frac{1}{5}$, $-\frac{1}{210}$ and $\frac{1}{1260}$ have been computed.

2. Using the divided differences from part (a), write the polynomial interpolant $p_3(x)$ of degree at most 3 in *Newton's nested form.*

3. Use the polynomial interpolation error theorem to find an upper bound for

$$\big|f(6) - p_3(6)\big|.$$

Question 2.

1. What is the definition of the Chebyshev polynomials of the first kind $T_n(x)$?

2. Derive the three-term recurrence relation for the Chebyshev polynomials T_n.

3. Give T_0 and T_1.

4. We are given
$$T_2(x) = 2x^2 - 1$$
and
$$T_3(x) = 4x^3 - 3x$$
Use the recurrence relation from part (b) to compute $T_4(x)$.

5. Find all the zeros of $T_4(x)$.

6. Sketch the graph of $T_4(x)$ on $[-1, 1]$. Clearly indicate its zeros and its values at the points where it attains 1 or -1.

7. State (without proof) the minimax property of the Chebyshev polynomial $\frac{1}{8}T_4$.